DPUTTSClientSNDX

Connects to a computer running DPUTTSServerPico and allows for electronic speech synthesis („text-to-speech“) using the Pico speech synthesis engine.

Audio data that is created that way will be played back on the computer running DPUTTSClientSNDX over the sound simulation system SNDX (i.e. OpenAL).

1. Inputs:

Text	Text that should be processed next. Pico markup commands can be used (see below).
Speak	Changing this input from 0 to 1 triggers synthesis of Text and playback.
Volume	Volume of the playback [0..1] (default: 1)
PositionMode	Positioning mode of the source 0: no positioning in the world, that is the playback will be at the position of SNDXListener. 1: absolute positioning in the SILAB 3D world. 2 (default): relative positioning to the SNDXListener
X, Y, Z	Position of the sound source
vX, vY, vZ	Velocity of the sound source relative to the corresponding axis. If no value is given (i.e. the inputs aren't connected) the velocity is calculated automatically by numerical derivation of the position.

2. Static parameters:

ServerIP	IP address or alias name of the computer where DPUTTSServer is running. default: *TTS*
Port	TCP port of that server that should be connected to. default: 0 which means the port given in the SILAB network configuration (*sys.par*) is used.

3. Dependencies:

DPUTTSServerPico must be running at the computer named at ServerIP.

DPUSNDXListener has to run on the same computer as *DPUTTSClientSNDX* to manage the SNDX sound system.

4. Functional principle:

DPUTTSServerPico opens a TCP port and waits for incoming connections.

As soon as a connection has been established, all data is read line by line, interpreted as UTF8 coded strings then synthesised using the Pico speech system and sent back to DPUTTSClientSNDX over the network.

DPUTTSClientSNDX works opens a connection to *DPUTTSServerPico* and receives the synthesized audio data over the network connection and plays it back over OpenAL. The audio source can be positioned in the 3D SILAB world.

DPUTTSSCNX may be loaded additionally to DPUTTSClient or DPUTTSClientSNDX and allows to trigger playback via events of the SCNX database. (See documentation for details.)

5. Example:

```
DPUTTSServerPico TTSPico
{
	Computer = {TTSP};
	Index = 20;
	TreeviewGroup = "Sound";
};

DPUTTSClientSNDX TTSClientPico
{
	Computer = {TTSP};
	Index = 50;
	TreeviewGroup = "Sound";
	ServerIP = "TTSP";
};

DPUSNDXListener listen
{
	Computer = {TTSP};
	TreeviewGroup = "Sound";
	Index = 20;
};

DPUTTSSCNX TTSSCNX
{
	Computer = {TTSP};
	TreeviewGroup = "Sound";
	Index = 51;
};
```

6. Pico markup commands:

Pico markup commands can be added to the text to modify speech output.

A complete description of the Pico markup commands can be found in the SVox Pico documentation. Some important ones are:

<pitch level='nnn'> … </pitch>	Sets the pitch of the voice to the given level. *nnn* may range from 50 to 200 where 100 marks the normal pitch level. Only text between the opening and clos-ing pitch tag is affected.
<speed level='nnn'> … </speed>	Sets the speed of the voice to the given level. *nnn* may range from 50 to 200 where 100 marks the normal speech speed. Only text between the opening and closing speed tag is affected.
<volume level='nnn'> … </volume>	Sets the volume to the given level. *nnn* may range from 50 to 200 where 100 marks the normal volume. Only text between the opening and closing volume tag is affected.
<phoneme ph='abc'/>	Outputs a word in phonetic script. This is useful in case the (uncommon) word is not pronounced correctly. abc is the SAMPA code of the word to speak. An introduction to SAMPA transcription is available on Wikipedia [at https://en.wikipedia.org/wiki/SAMPA_chart in English and at https://de.wikipedia.org/wiki/SAMPA-Transkribierungscodes in German]. A com-plete description of the supported phonetic codes for each language is available in the SVox Pico manual [available at https://salsa.debian.org/a11y-team/svox/blob/debian-sid/pico_resources/docs/SVOX_Pico_Manual.pdf].